QUIZ 1

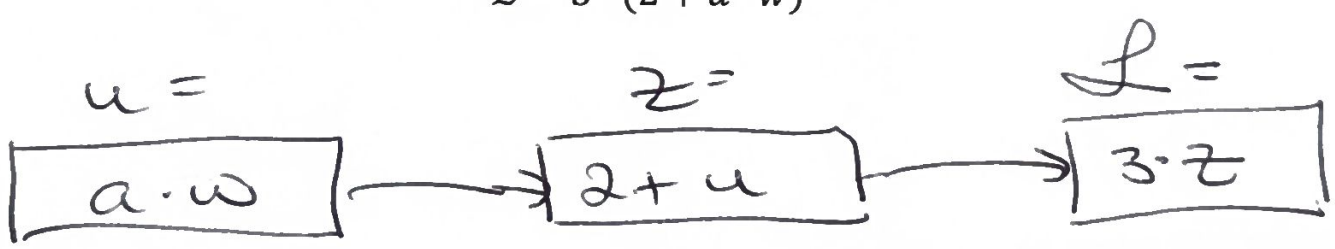
1. **Deep Learning.** In 1-2 sentences, explain the key difference between 'shallow' learning and 'deep' learning that we discussed in class.

→ Shallow learning refers to ML w̄ few transformations / representations of data. This refers to most traditional ML for supervised learning, e.g. XGBoost, Logistic, SVM, etc

→ "Deep" learning is made possible w̄ neural nets. Many hidden/intermediate layers allows for many transformations to be applied.

2. **Backpropagation 1.** Considering the following function, draw the associated computation graph. Use the computation graph to figure out the gradient/derivative of $\mathcal{L}$ with respect to w, or $d\mathcal{L}/dw$ (that is, show how $\mathcal{L}$ will change with respect to w by tracing derivatives through the computation graph). Note that the current value of a is 5.

$$\mathcal{L} = 3 \cdot (2 + a \cdot w)$$



$$\partial\mathcal{L}/\partial w = \frac{\partial \mathcal{L}}{\partial z} \cdot \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial w}$$

$$= 3 \cdot 1 \cdot \cancel{a} \rightarrow 5$$

$$= 15$$

3. **Backpropagation 2.** Given the derivative you calculated above, a learning rate of 0.01, and a current value of 0.5 for w, what would we update w to in the current 'training step', to minimize $\mathcal{L}$?

$$w_{new} = w_{old} + (-1) \cdot \left(\frac{\partial \mathcal{L}}{\partial w}\right) \cdot \alpha.$$

$$= 0.5 - 15 \cdot 0.01$$

$$= 0.5 - 0.15$$

$$= 0.35$$

4. **Output Activation.** Explain in your own words why it would be incorrect to employ a sigmoid activation function in the final (output) layer of a neural network that is intended to predict household income.

→ Sigmoid can only produce output on the range (0,1). Household income is generally >> 1 (in dollars).

5. **Loss Function.** Which of the following loss functions would make the most sense for a binary label / binary prediction problem?

A) Mean squared error
B) Cross-entropy / log loss
C) L1 loss → MAE
D) L2 loss → MSE

Answer: B